Annex A

Computational Thinking Exercise: "Smart School Canteen Queue"

Section: 9 -Potassium **Score:**____________ **C# /**
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Scenario
The PSHS school canteen is small and often gets crowded during lunch break. Students line up to buy food, but the process is slow because:

- Some students take too long to decide what to order.
- The cashier has to manually calculate totals and give change.
- There is no system to track which food items are running out.

Your group's task is to **decompose this problem** into smaller, manageable parts that could be solved with computational thinking (CT) Skills.

Step 1: Identify the Big Problem
Main Problem:
The school canteen manual, slow ordering and payment process cause long ques, causing for the canteen to crowd.

Step 2: Identify three to four Sub-Problems
Please list possible sub-problems:

1. **Students take too long deciding what to order, slowing down the line.**

2. **The cashier manually calculates totals and change, which is slow and can cause error.**

3. **There's no way to track which food items are running low or sold out, so students waste time ordering unavailable items.**

4. **There's no way to manage the flow of students so the line moves in an messy way.**

Step 3: Define Computational Thinking Approaches
For each sub-problem, apply CT skills:

Sub-Problem	CT Skill	Example Solution
1	Abstraction	Give labels onto the foods and group them in simple categories.
2	Algorithm Design	The cashiers can use calculator app from their phones or the calculators.

3	**Pattern Recognition**	**Track daily/weekly sales data to further notice which items sell out.**
4	**Decomposition**	**Break the canteen process into separate stations each with a different function.**

Step 4: Draw a flowchart or write a pseudocode for the identified sub-problem

FLOWCHART IS AT BOTTOM PAGE

Rubrics For Grading

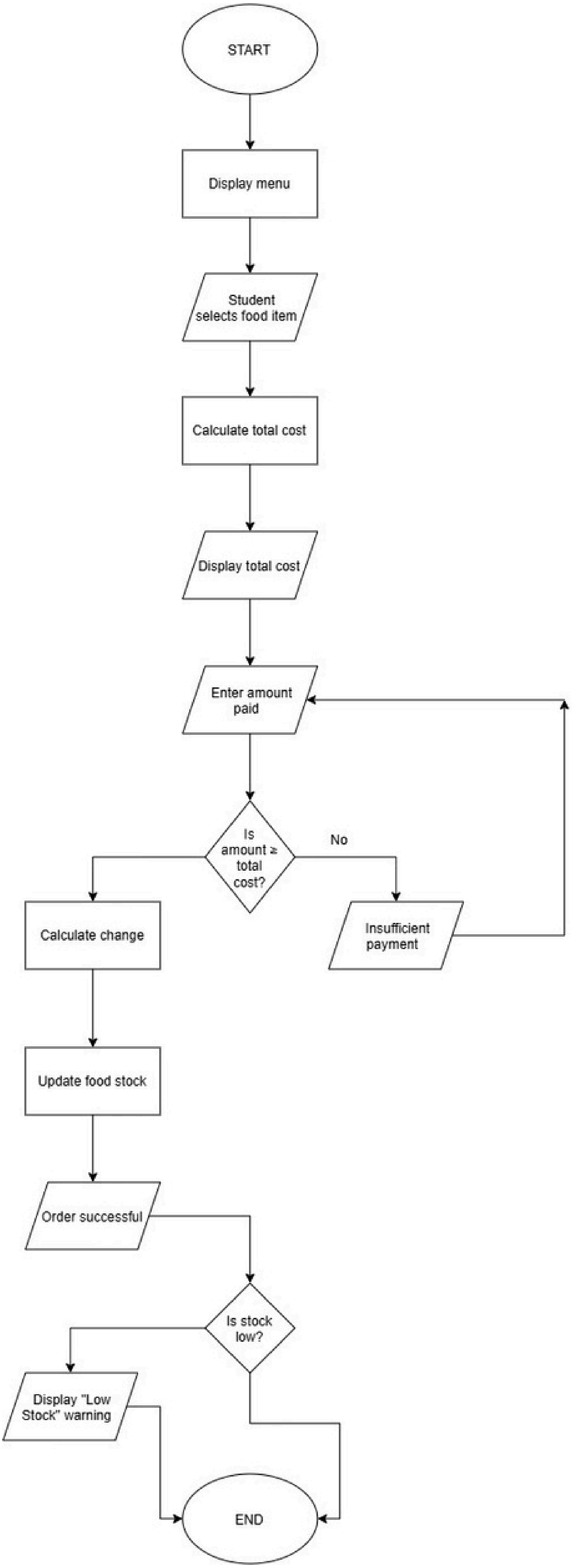
Total Points: 20pts

Criteria & Levels of Performance

Criteria	Excellent (4)	Good (3)	Fair (2)	N.I. (1)
Identificati on of Sub-Proble m s	Identifies 3+ clear, relevant sub-problems that directly connect to the scenario.	Identifies 2–3 mostly relevantsub-problems.	Identifies 1–2 vague or partially relevant sub-problems.	Struggles to identify sub-proble ms or lists unrelated issues.

Application ofCT Strategies	Correctly applies appropriate CT strategies (abstraction, decomposition, pattern recognition, algorithm design) to each sub-problem with clear reasoning.	Applies CT strategies to most sub-proble ms, with minor errors or limited explanation.	Applies CT strategies inconsisten tly, with weak or unclear reasoning.	Rarely appliesCT strategies or misuses them.

Flowchart / Pseudocode X 2	Flowchart / Pseudocode is complete, logical, and easy to follow; shows clear steps and decision points.	Flowchart / Pseudocode ismostly complete and logical, with minor gaps or unclear steps.	Flowchart / Pseudo Code ispartially complete, missing key steps or connections.	Flowchart / Pseudocode is incomplete, confusing, or missing entirely.
Reflection / Explanation	Provides thoughtful reflection on how decomposition helps problem-solving and identifies CT skills used with strong justification.	Provides adequate reflection with some justification of CT skills.	Provides limitedreflection with weak or genericjustification.	Provides minimal or no reflection.


START
Display menu
Student selects food item
Calculate total cost
Display total cost
Enter amount paid
Is amount ≥ total cost?
No
Calculate change
Insufficient payment
Update food stock
Order successful
Is stock low?
Display "Low Stock" warning
END